

Module 2

Routing and Addressing at Internet Scale

1 · Intra-AS routing protocols

2 · Inter-AS routing and AS policy

3 · Aggregation, NAT and IPv6

A quick recap of routing protocols

We can categorise a routing protocol by the **algorithm** underneath it — in Module 1 we met **distance vector** and **link state**.

We can also categorise it by **where it runs** — inside one autonomous system, or between them.

algorithm ↓ domain →	Intra-AS within one autonomous system	Inter-AS between autonomous systems
distance vector	RIP Routing Information Protocol now	
link state	OSPF Open Shortest Path First now	
path vector a close relative of distance vector		BGP Border Gateway Protocol next



MODULE 2 · ROUTING INSIDE ONE ADMINISTRATIVE DOMAIN

Intra-AS Routing Protocols

1 · Intra-AS routing protocols

2 · Inter-AS routing and AS policy

3 · Aggregation, NAT and IPv6

An overview of intra-AS routing protocols

RIP

Routing Information Protocol

DISTANCE VECTOR

this lecture

OSPF

Open Shortest Path First

LINK STATE

this lecture

IS-IS

Intermediate System to Intermediate System

LINK STATE

A link-state alternative to OSPF, widely used inside ISP networks. It works so much like OSPF that we will not cover it separately.

A quick recap of the distance-vector routing protocol

- 1 Each router keeps a **distance vector** — its own best known cost to every destination, and the **next hop** on that path.
↓
- 2 Each router sends its vector **only to its direct neighbours**, and receives theirs in return. No router ever learns the full topology.
↓
- 3 On a neighbour's vector the router applies **relaxation** — if that neighbour offers a cheaper path, take it: update the cost and make that neighbour the next hop. Any entry that changed is advertised onward.

A closer look at RIP

1 RIP is a distance-vector protocol

It runs the algorithm we have just recapped, within a single AS. But how does RIP measure the **cost** of each path?

2 Every link costs one hop

RIP measures the cost of a path by its number of **hops**. In the graph from Module 1, that is simply giving **every edge weight 1**.



distance from A to D = 3 hops

3 15 hops is as far as RIP counts

A path of more than **15 hops** is treated as unreachable — a metric of 16 means infinity.

Why impose a limit at all? To bound count to infinity — we come back to this shortly.

THE TABLE THE DATAGRAMS USE, AND RIP'S OWN DATABASE

A RIP router keeps two tables

RIP routing table at A

destination	metric	next hop	timer
198.51.100.0/24	1	B	180 s
203.0.113.0/24	1	E	180 s
192.0.2.0/24	2	B	180 s

Forwarding table at A

destination	next hop
198.51.100.0/24	B
203.0.113.0/24	E
192.0.2.0/24	B

RIP ROUTING TABLE

To run distance vector, every RIP router keeps a **second table** of its own — its **RIP routing table**.

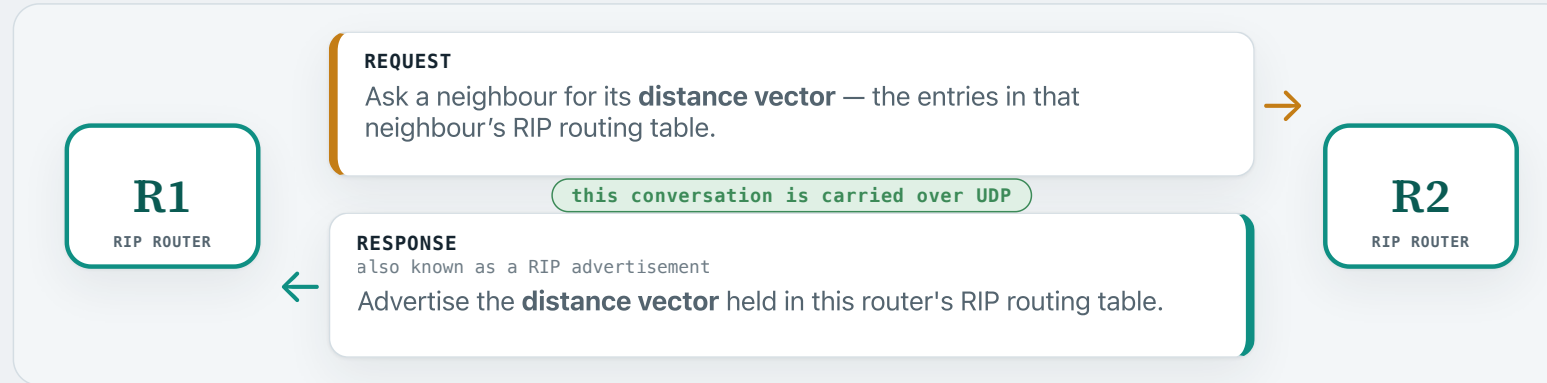
For every destination it records more than the forwarding table does:

- the **metric** — how far away that destination is, its entry in the **distance vector**
- the **next hop** on that path
- a **timer** — RIP's own bookkeeping, giving the entry its time to live

RIP routing-table entries and timers: RFC 2453 §§3.5, 3.8

The RIP conversation: Request and Response

Distance vector rests on routers **talking to their direct neighbours**. So how does RIP carry that conversation?



AFTER A REQUEST

Requested

Return all routes or the selected routing-table entries requested by another router.

ROUGHLY EVERY 30 SECONDS

Periodic

Send the current routing table to neighbouring RIP routers.

AFTER A ROUTE CHANGES

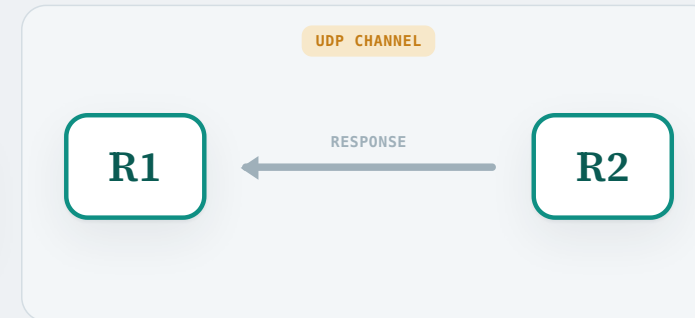
Triggered

Send changed routing information shortly after the change occurs.

An example of running RIP

RIP routing table at R1

destination	metric	next hop	timer
203.0.113.0/24	16	R2	expired



7 · What if R2 stops sending them?

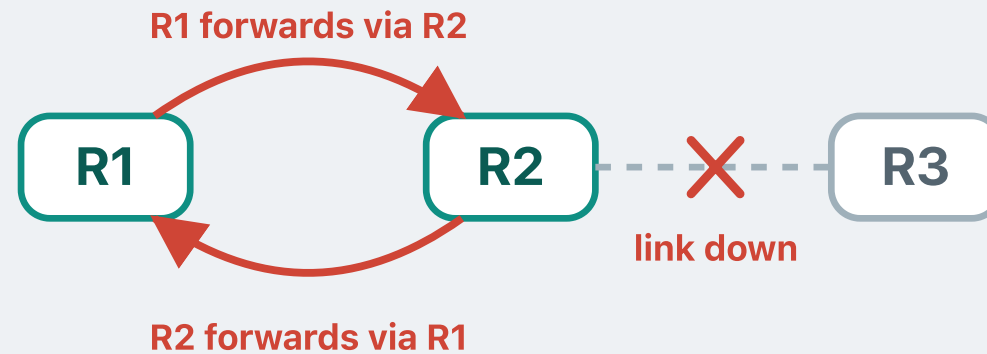
- **The timer runs out.** No valid refresh arrives from R2, and after 180 seconds the entry expires.
- **Mark it unreachable.** R1 sets the cost metric to **16** — RIP's infinity — and immediately drops the route from its **forwarding table**.
- **Keep it, to tell the neighbours.** The entry stays in the RIP routing table for another **120 seconds**, advertised with metric 16 so the neighbours learn the destination is gone.
- **Then remove it.** After those 120 seconds, R1 deletes the entry from its routing table altogether.

Why a distance vector can count to infinity

A distance-vector protocol suffers from the **count-to-infinity problem**.

The cause is what a distance vector carries: **distances, not paths**. A neighbour is told *how far*, never *through whom* — so a router cannot see that the route it is offered runs back through itself.

Suppose the link onward to **R3** fails. R2 hears that R1 can still reach R3, while R1's own route to R3 runs through R2 — so each forwards to the other, and their costs climb together, one hop at a time.



How RIP limits count to infinity

POISON REVERSE

Send it back as infinity

If R1's route to a destination runs through R2, R1 advertises that destination back to R2 at a cost of **infinity** — so R2 never tries to reach it through R1.

Only a **heuristic**. It breaks a loop between **two** routers, but not one running through three or more.

COUNT ONLY TO 16

Make infinity deliberately small

So the counting cannot run away for long: at **16** a route is already infinity. That is why the maximum path length in RIP is **15 hops**.

This is RIP's **fundamental solution** to the count-to-infinity problem.

OSPF · Open Shortest Path First

What it is. OSPF is an intra-AS routing protocol which uses the **link-state** routing algorithm.

Protocol channel. OSPF carries its messages **directly over IP**, without TCP or UDP.

What they carry. Those messages carry the **link-state advertisements** — so before going further, a reminder of how link state works.

A quick recap of the link-state routing protocol

- 1 Each router creates an **advertisement** for its local information, and tells the others.
- ↓
- 2 Every router uses all the advertisements to build the **same topology**.
- ↓
- 3 Each router runs a **shortest-path algorithm** over that topology to derive its forwarding table.

OSPF provides useful features for a large AS

Authentication

- Routers accept updates only from configured participants.
- Without it, an **attacker** on the segment could advertise a **cost-1 path** to a prefix and pull that traffic through itself — to drop it, or to read it.

Operator-defined costs

- The operator sets each link's cost, and so decides which paths OSPF prefers.
- Costs can follow **bandwidth**, so a fast link beats a slow one.

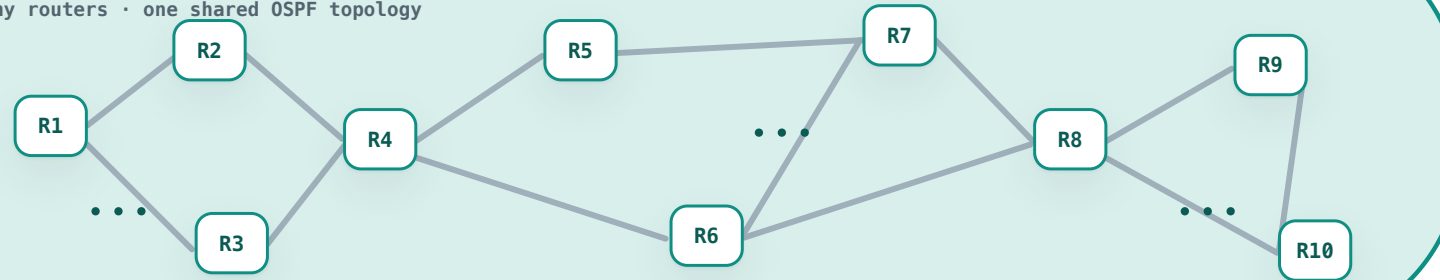
Area routing

A large AS can be divided into **areas**, so that OSPF scales.

Why can a large AS strain OSPF?

One large AS

many routers · one shared OSPF topology



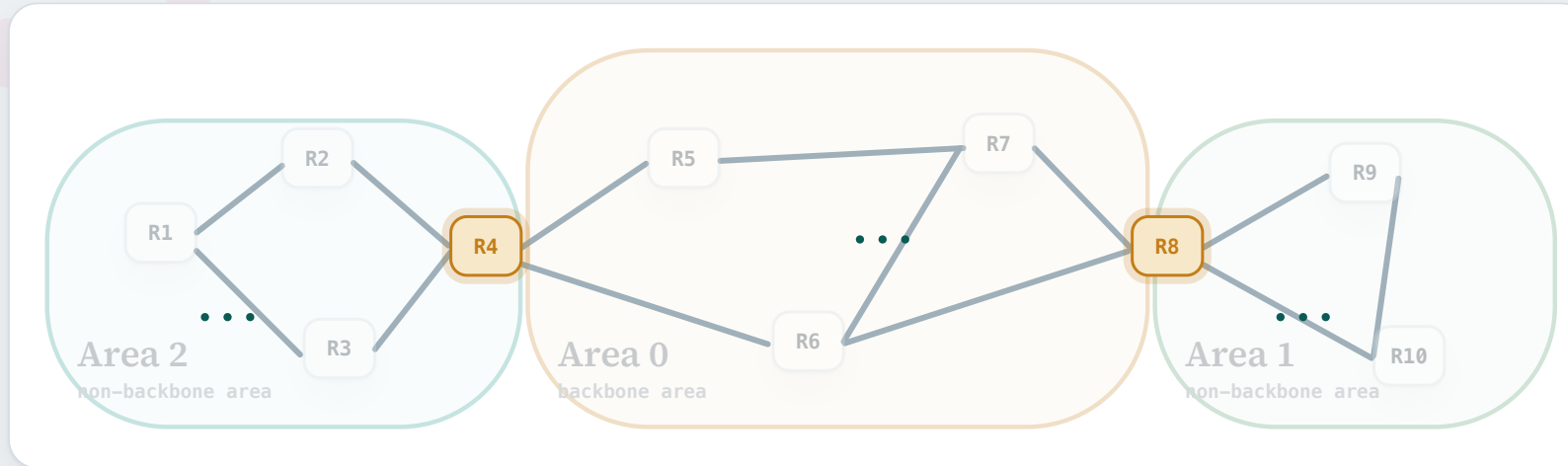
More memory

Every router stores the complete AS topology in its link-state database.

Slower convergence

Every change must be flooded widely before all routers calculate the new topology.

OSPF divides one AS into smaller areas



Non-backbone areas

Areas 1 and 2 each contain a smaller group of routers and networks.

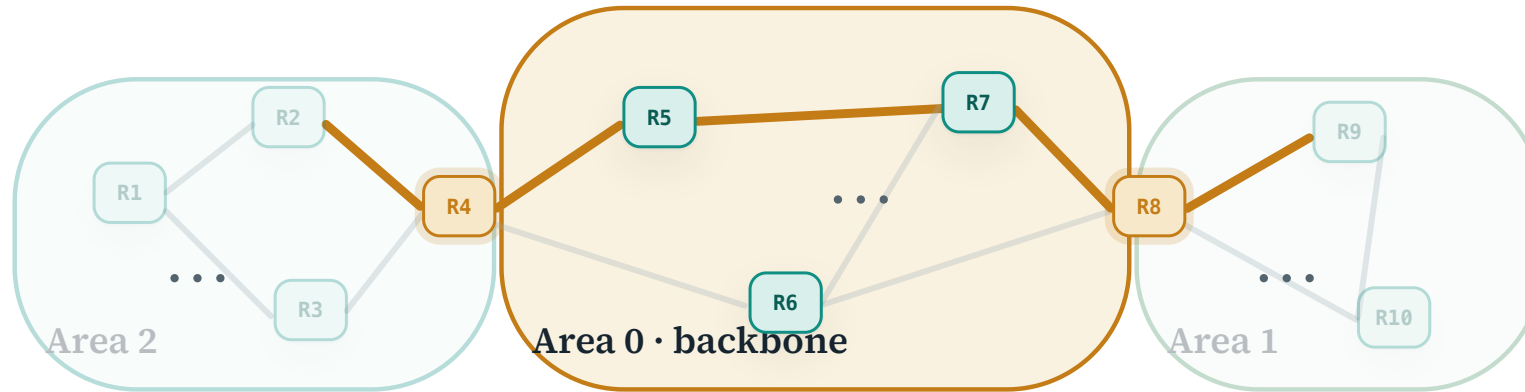
Backbone area

Area 0 connects the non-backbone areas and carries inter-area reachability.

Area border routers

R4 and R8 each participate in Area 0 and one non-backbone area.

How does hierarchical OSPF fit together?



Inside an area

- A router does its link-state flooding only within its own area, and stores that area's topology.
- Each router then runs the shortest-path algorithm over that topology to reach destinations inside its area.

Between areas

- ABRs carry condensed inter-area reachability through the backbone, Area 0.
- For example, a route from Area 2 to Area 1 can follow:
Area 2 → R4 → Area 0 → R8 → Area 1

OSPF areas and inter-area routing: RFC 2328 §§3, 3.1–3.3, 4.1

Intra-AS routing recap

● RIP is distance vector.

- It measures distance in router hops, and caps a path at 15.
- Neighbours exchange distance vectors, kept alive by periodic Responses.

● OSPF is link state.

- Routers flood advertisements, so each builds the same topology.
- A large AS is divided into areas so that OSPF scales.



MODULE 2 · ROUTING BETWEEN ADMINISTRATIVE DOMAINS

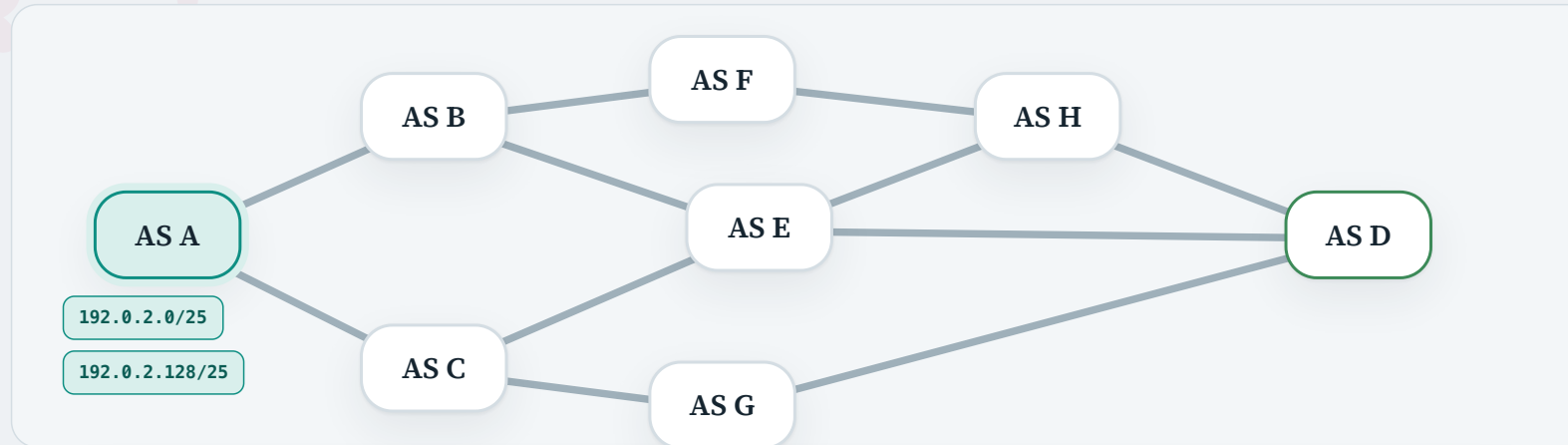
Inter-AS Routing and AS Policy

1 · Intra-AS routing protocols

2 · Inter-AS routing and AS policy

3 · Aggregation, NAT and IPv6

The Internet is a network of autonomous systems



SO FAR, INSIDE ONE AS

- RIP and OSPF both run **within a single AS** — many routers forming **several subnets**, all managed by one organisation.
- Each subnet is identified by a **network prefix**, written in **CIDR** notation — AS A holds 192.0.2.0/25 and 192.0.2.128/25.
- So 192.0.2.1 sending to 192.0.2.200 stays **inside AS A**: that is **intra-AS routing**.

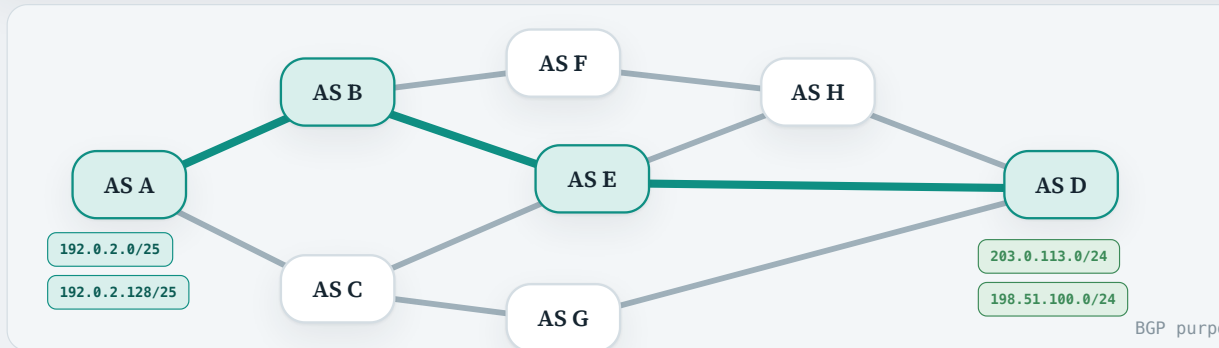
BGP is the Internet's inter-AS routing protocol

BGP is the Internet's answer to that problem. Here is how it does it.

LEARN EXTERNAL REACHABILITY

- Route-finding over a graph of **nodes** and **edges**, just as in Module 1 — except a node is no longer a router but a whole **autonomous system**, holding one or more subnets.
- Unlike Module 1, a destination is not a single IP address. It is a **destination prefix** in CIDR notation, like **203.0.113.0/24**, identifying a subnet inside some AS. **Every AS advertises its own prefixes** this way.
- So for each **destination prefix**, an AS must find a path to it across that graph — a path made of several ASes.

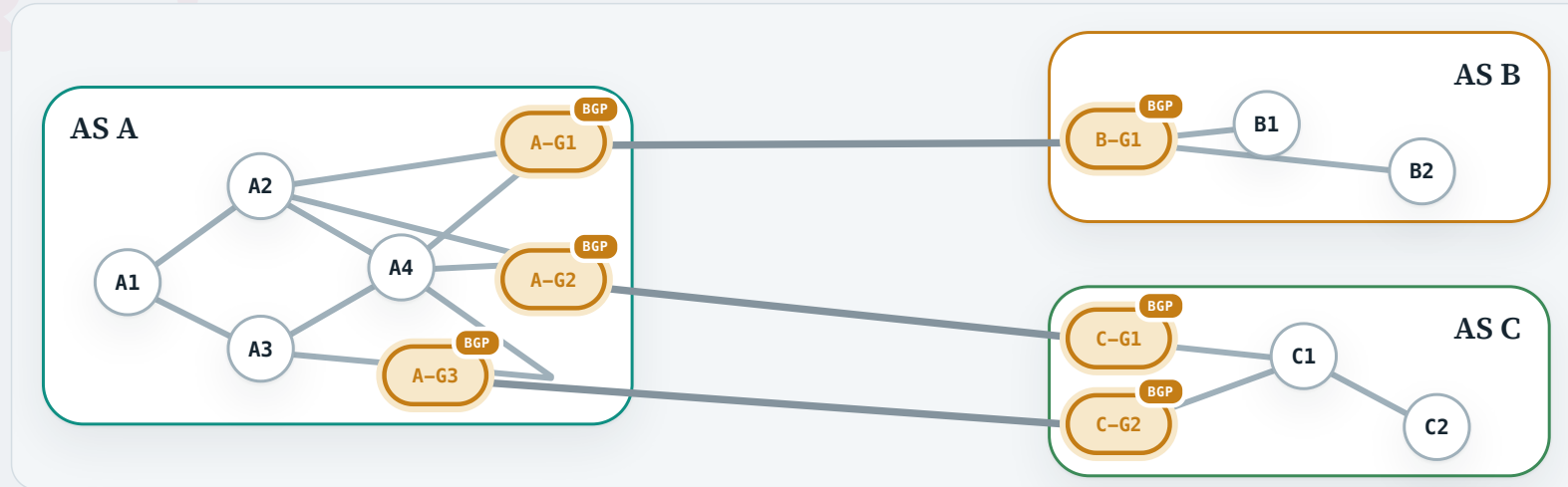
e.g. a path from AS A to AS D: **AS A → AS B → AS E → AS D**



BGP purpose and operation: RFC 4271 §§1, 3, 9.1

WHO PARTICIPATES IN BGP?

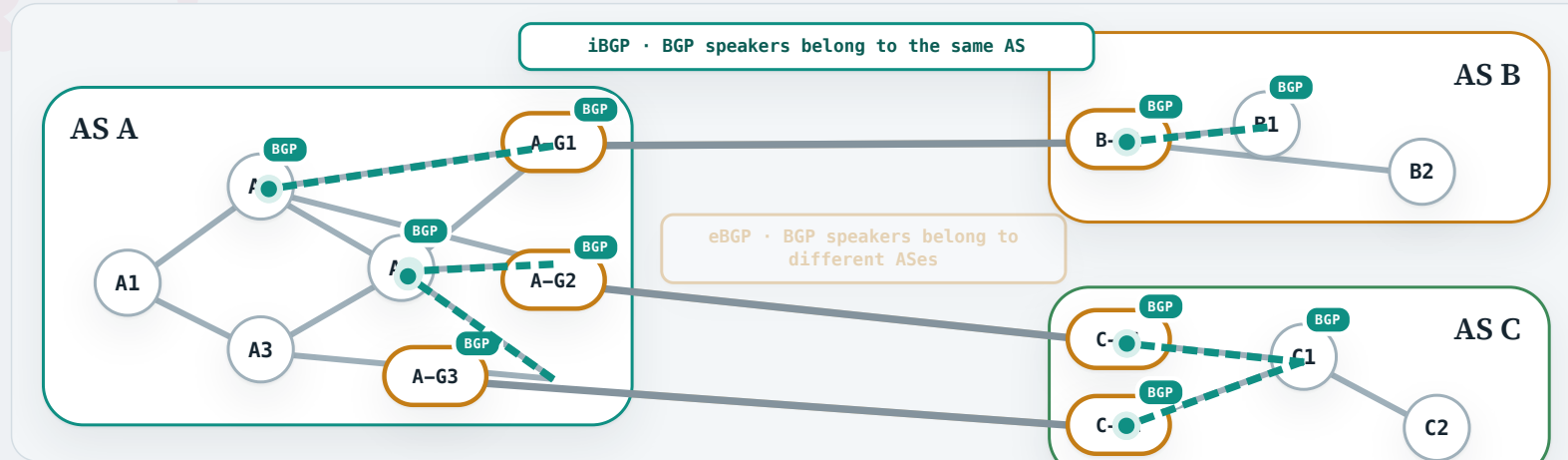
BGP is run by BGP-speaking routers



USUALLY AT THE EDGE

A **border** or **gateway** router is one that links directly to a router in another AS.
Such routers normally run BGP — here, all six gateways are BGP speakers.

A BGP session is external or internal



External BGP (eBGP)

— this is how an AS **learns external reachability**

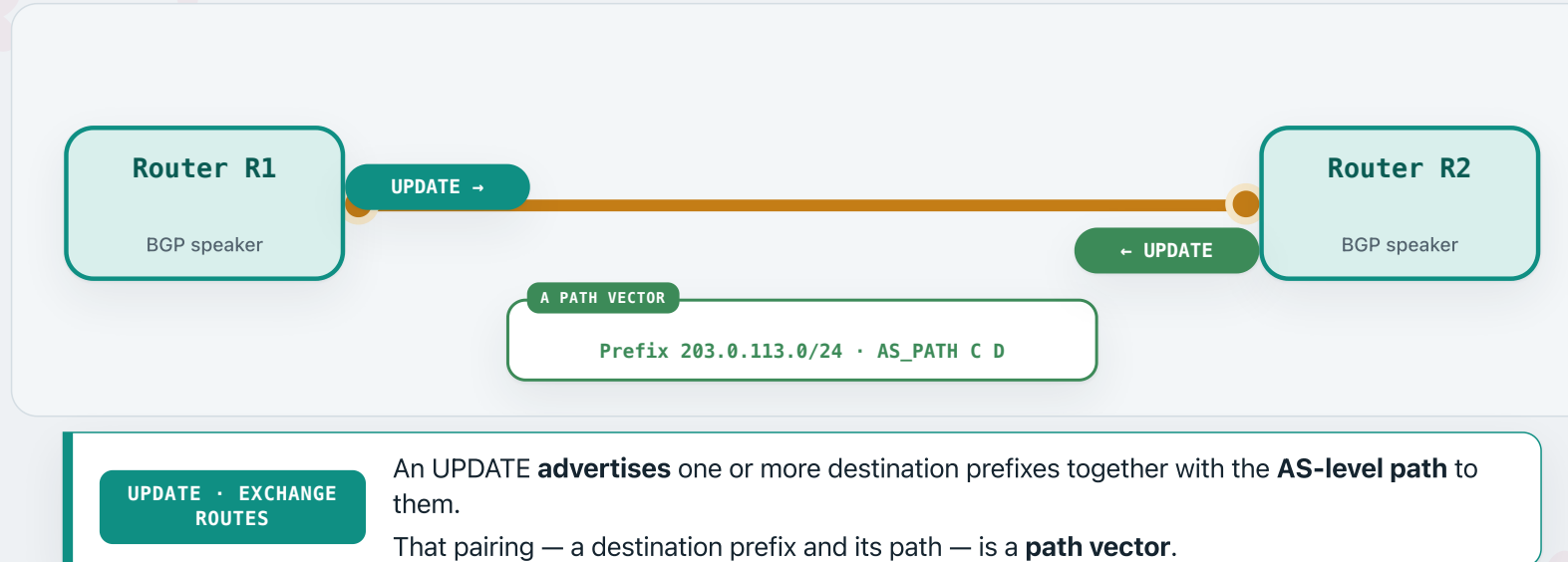
Sessions between BGP speakers in **different ASes**, usually the border speakers each side of the boundary. Using **eBGP**, an AS learns **AS-level paths** to destinations outside itself.

Internal BGP (iBGP)

— this is how the AS **distributes it inside**

Sessions between BGP speakers in the **same AS**. Using **iBGP**, all the routers inside the AS come to know those external paths, so each of them can forward towards the outside.

A BGP session is a structured conversation



A closer look at a BGP path vector

Unlike a distance-vector protocol, what BGP advertises to a neighbour is a path vector.

ONE PATH-VECTOR ROUTE

NLRI 203.0.113.0/24
AS_PATH AS E · AS H · AS D
NEXT_HOP E1

NLRI

Network Layer Reachability Information: the destination prefix being advertised.

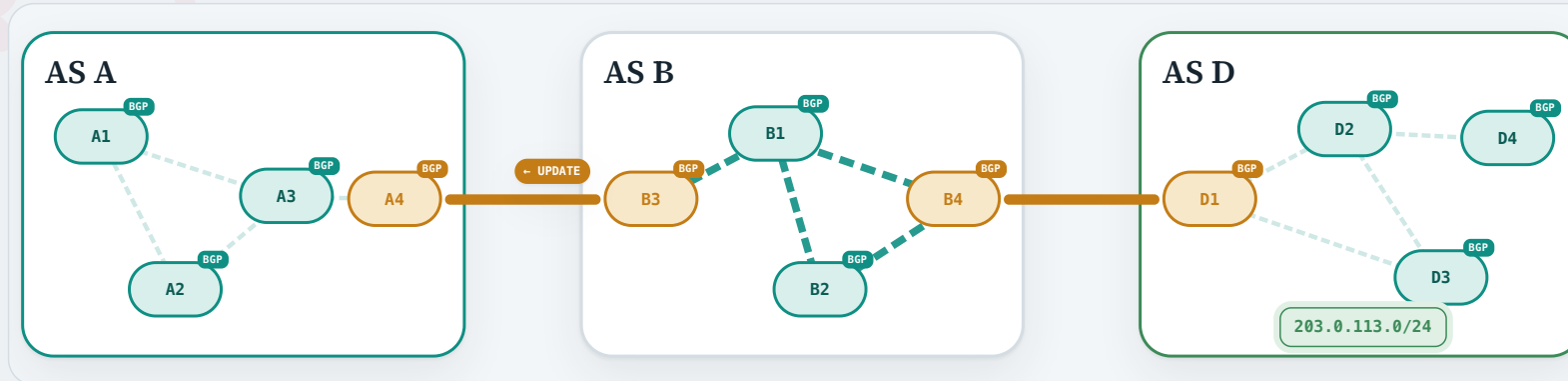
AS_PATH

The complete **AS-level path** leading to that destination prefix.

NEXT_HOP

The IP address of the **gateway router** to send traffic to for that prefix.

Watch one BGP route travel across three ASes



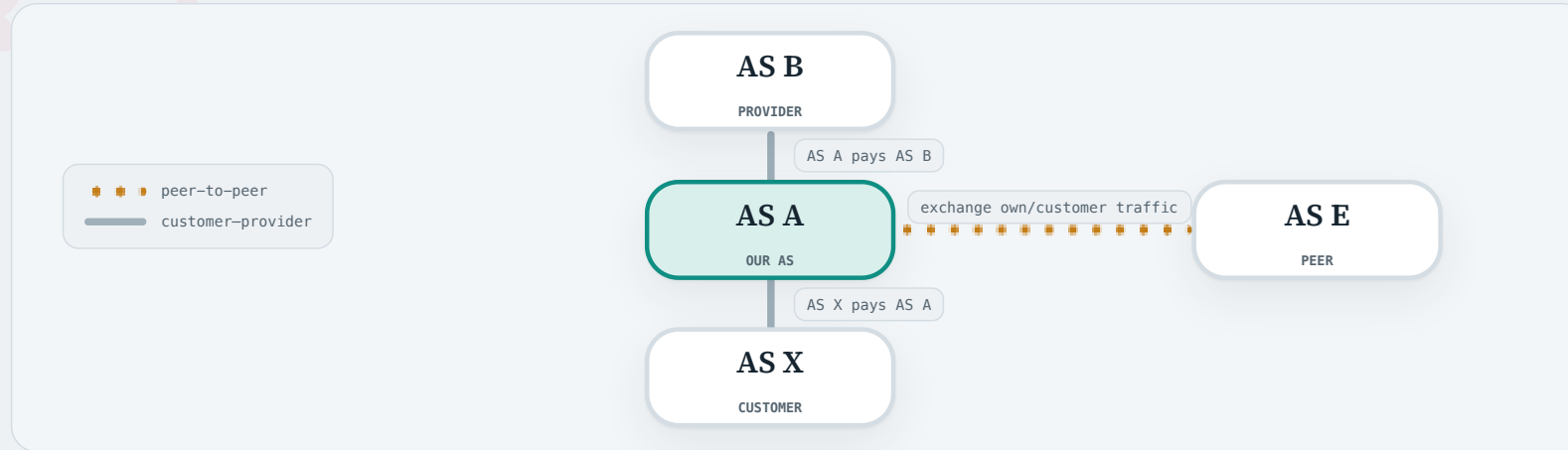
4 · EBGP: B → A

- AS A is a **neighbour** of AS B.
- Suppose AS B is willing to export what it has just learned to AS A.
- B3 **prepends AS B** onto the AS_PATH and sets NEXT_HOP to its own address.
- It sends this new path vector in an **UPDATE** to AS A's border router A4.

```
NLRI      203.0.113.0/24
AS_PATH   AS B · AS D
NEXT_HOP   B3
```

BGP UPDATE, path attributes, and import/export policy: RFC 4271 §§3, 4.3, 5.1.2–5.1.3, 9.1–9.2

Business relationships between ASes



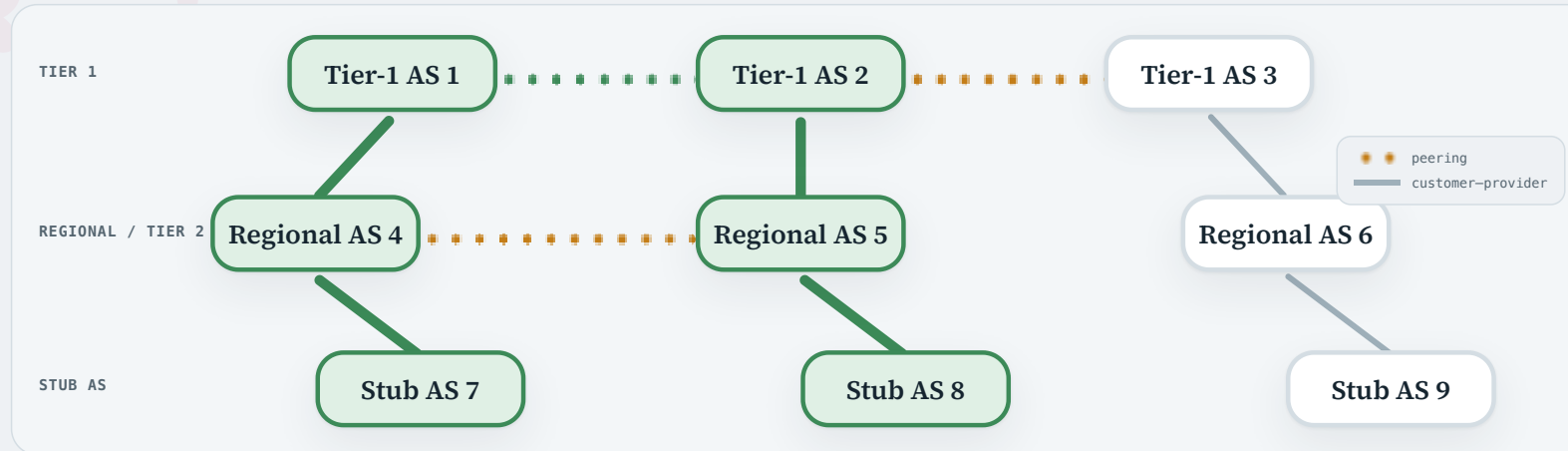
THE QUESTION

What drives an AS's import and export policy?

The answer lies in the **business relationships** between ASes — who pays whom for carrying traffic.

HOW CUSTOMER, PROVIDER, AND PEER RELATIONSHIPS REPEAT ACROSS THE INTERNET

These relationships create a commercial AS hierarchy



VALLEY-FREE PATHS

Whenever every AS's import and export policy is guided by these relationships, a path takes an **up-across-down** shape — a **valley-free path**.
e.g. a datagram from stub AS 7 to stub AS 8 climbs through providers to the Tier-1 level, crosses one peer link, then descends to AS 8.

Simplified AS hierarchy and valley-free path model: CAIDA AS Relationships; Gao, 2001

A quick recap of BGP so far

- **There are two kinds of routing.**
 - Inside a single AS it is **intra-AS routing** — RIP, OSPF.
 - Between ASes it is **inter-AS routing**, and **BGP** is the Internet's answer to it.
- **BGP has two halves.**
 - **eBGP** runs between ASes, and learns the **AS-level path** to prefixes in other ASes.
 - **iBGP** runs inside one AS, and carries what was learned to **every BGP speaker** in it.
- **Choosing a route is not one decision.**
 - What an AS keeps and what it advertises are its **import and export policy**.
 - And those are driven by the **business relationships** we have just met — customer, peer, provider.

Import policy: which route should the AS prefer?

ROUTE LEARNED FROM	COMMON PREFERENCE	ECONOMIC INTUITION
Customer	HIGH	The customer pays us for transit.
Peer	MEDIUM	Usually settlement-free exchange.
Provider	LOW	We pay the provider for transit.

3 · PROVIDER ROUTE

Provider-learned routes commonly receive low preference because the AS must pay the provider to carry that traffic.

A COMMON COMMERCIAL GUIDELINE—NOT A RULE ENFORCED BY BGP

Export policy: where should the route be advertised?

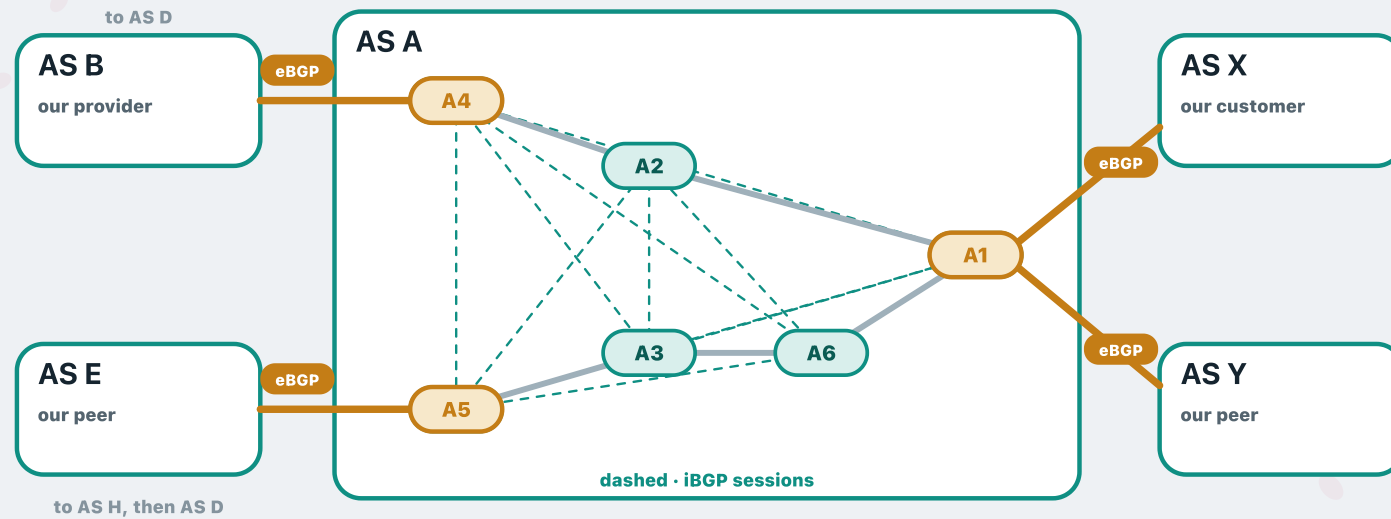
ROUTE LEARNED FROM	TO CUSTOMER	TO PEER	TO PROVIDER
Customer	✓	✓	✓
Peer	✓	✗	✗
Provider	✓	✗	✗

PEER/PROVIDER ROUTES

Do not normally export these to another peer or provider; doing so creates unpaid transit.

ZOOMING IN ON THE AS THAT RECEIVES

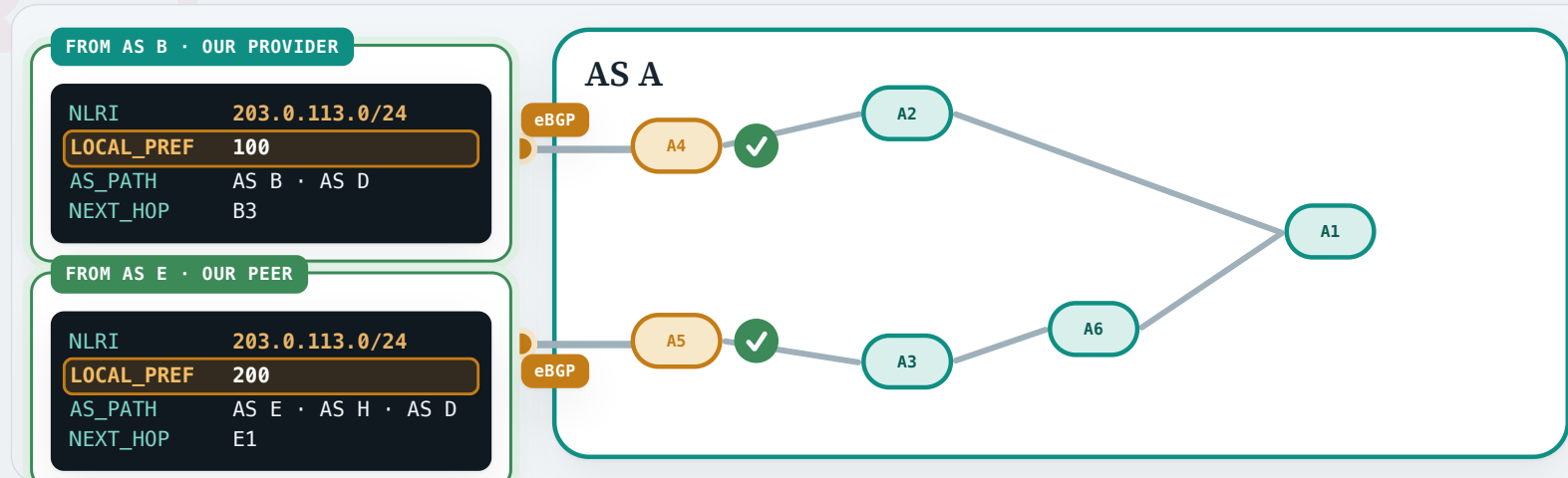
One AS, two ways in, two ways out



THE SETTING

- Now suppose we have the following destination: `203.0.113.0/24`, a subnet inside **AS D**.
- Suppose **both AS B and AS E can reach it** — and both would like to advertise this destination to AS A.

AS A applies its eBGP import policy



2 · IT ACCEPTS BOTH, AND RANKS THEM

Recall the import rule — **customer over peer over provider**. AS E is a peer and AS B a provider, so AS A prefers the **AS E** route.

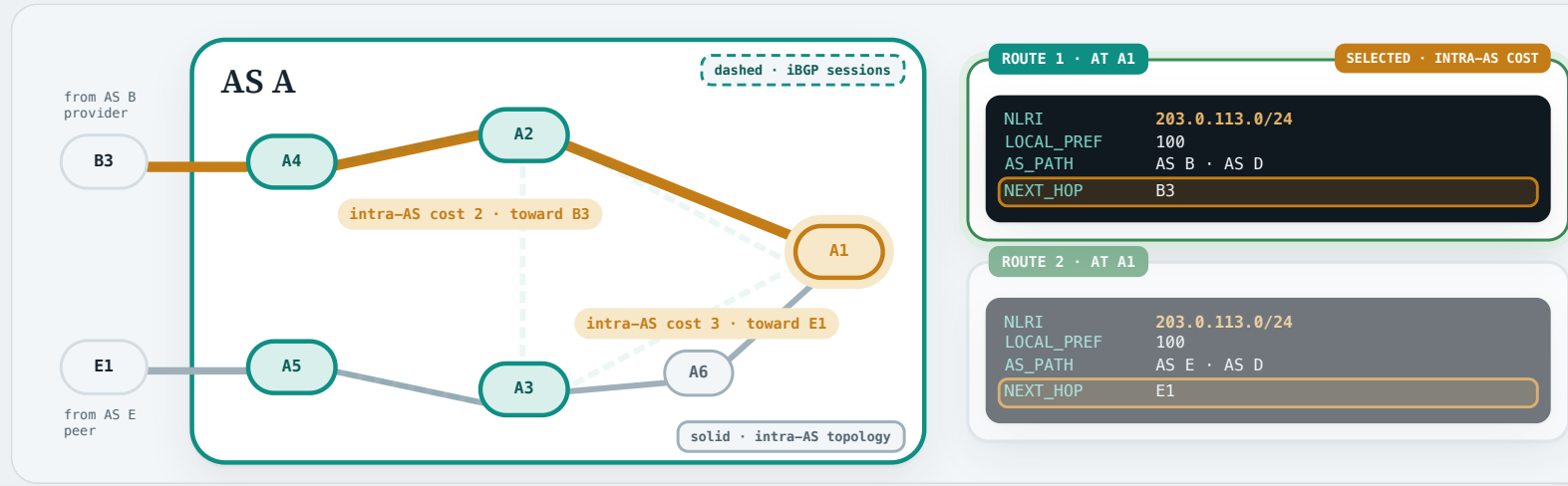
How is that preference recorded? In a LOCAL_PREF field on the path vector — **higher is better**.

So AS A accepts **both**, and sets LOCAL_PREF **200** on the AS E route and **100** on the AS B route.

Why keep the loser? As a **backup** if the preferred route is withdrawn.

ONE NEXT HOP PER PREFIX · LOCAL_PREF → AS_PATH → INTRA-AS COST

How does a single BGP router break a tie between routes?

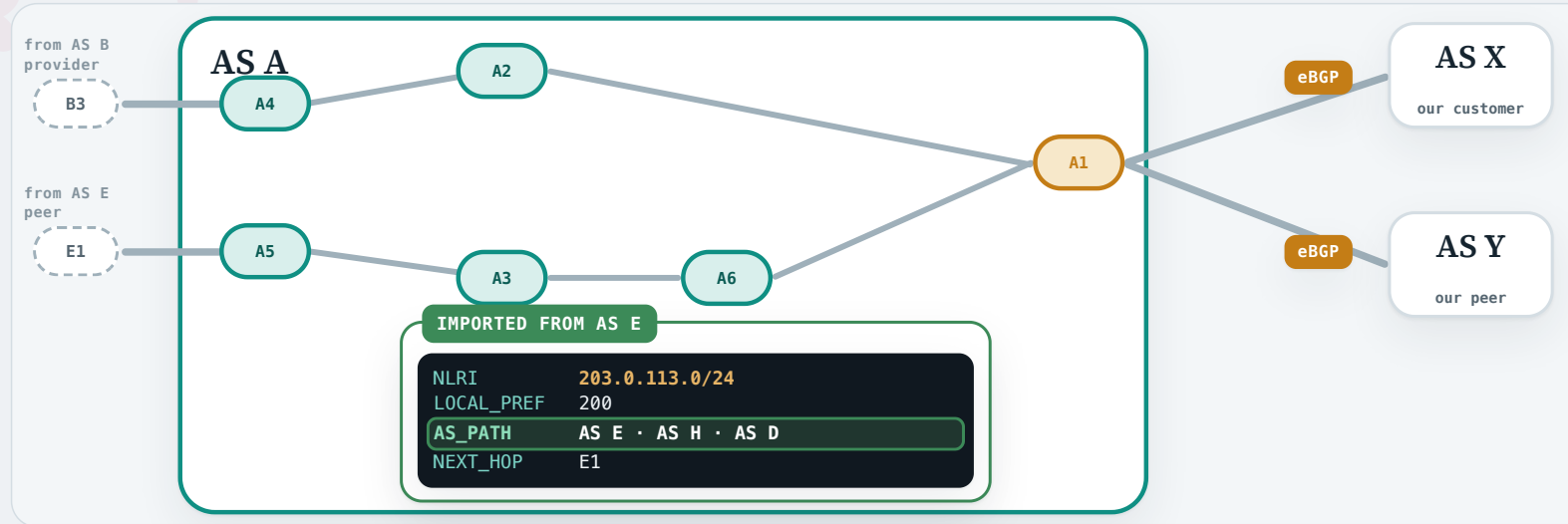


3 · THEN HOT-POTATO ROUTING

If both tie, A1 asks its own intra-AS routing protocol which next hop is closer: B3 at cost 2 beats E1 at cost 3, so **Route 1 wins — hot-potato routing**.

iBGP, distributed selection, LOCAL_PREF, and intra-AS cost to NEXT_HOP: RFC 4271 §§3, 5.1.5, 9.1.1–9.1.2

AS A applies its eBGP export policy



1 · THE SETTING

- Suppose A1 selected the route through AS E: **AS E · AS H · AS D**.
- A1 is a border router, so it can pass what it knows on to its neighbouring ASes — **AS X**, our customer, and **AS Y**, another peer.
- **To whom should A1 export this route?** That is what **export policy** decides.



MODULE 2 · KEEPING THE INTERNET ADDRESSABLE

IP Addressing Scaling: Aggregation, NAT and IPv6

1 · Intra-AS routing protocols

2 · Inter-AS routing and AS policy

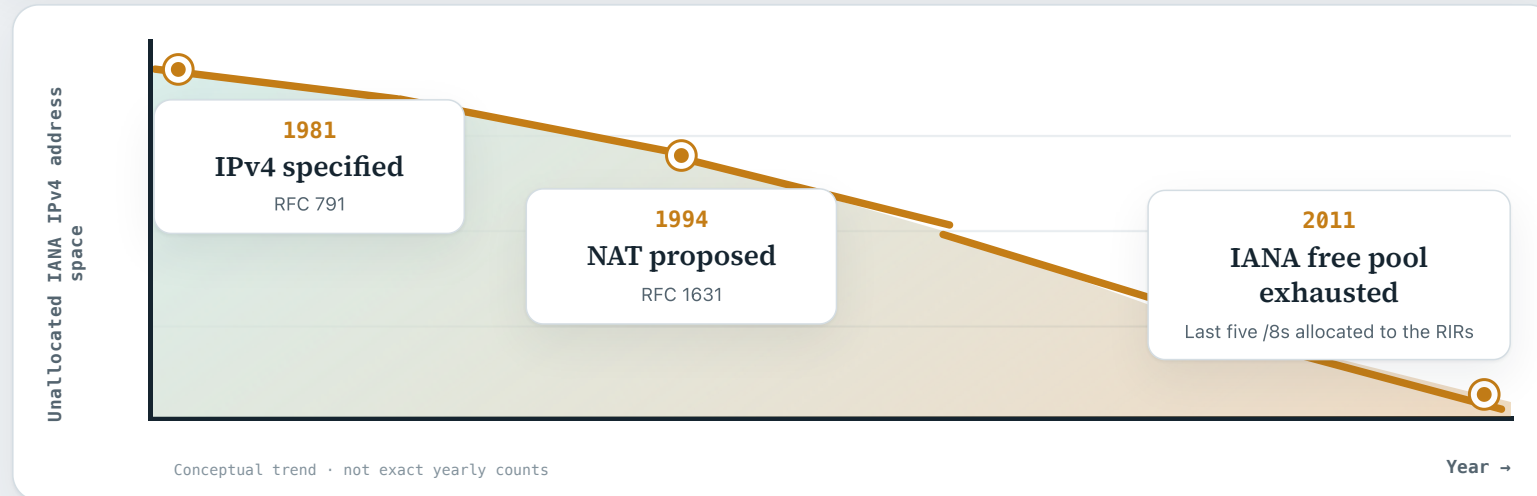
3 · Aggregation, NAT and IPv6

A BRIEF HISTORY

IPv4's address space became scarce as the Internet grew

2^{32} addresses

IPv4 uses 32-bit addresses. Its finite address space became an important limitation as more networks and devices joined the Internet.



RFC Editor: RFCs 791 and 1631 · IANA central-pool exhaustion: ICANN, 3 February 2011

How did the Internet adapt to IPv4 scarcity?

We will study three responses that extended IPv4's useful life and kept the Internet growing.

1

Classless Inter-Domain Routing

already covered in Term 3

2

Share public IPv4 addresses

Network Address Translation (NAT)

3

Use a 128-bit address space

IPv6

CIDR stopped IPv4 address space going to waste

20 bits

the university's block

3

department

9 bits

hosts inside that department

department	those 3 bits	its block
Engineering	000	200.23.16.0/23
Science	001	200.23.18.0/23
Law	010	200.23.20.0/23
Arts	011	200.23.22.0/23
...	...	...
Library	111	200.23.30.0/23

THE ANSWER

- Under **CIDR** the split can fall at **any bit**.
- e.g. a university takes **200.23.16.0/20** and divides it into eight sub-blocks: **3 bits** to say which department, and the remaining **9 bits** to say which host inside it.

IPv4 address aggregation in the forwarding table

A ROUTER'S FORWARDING TABLE

network	destination	next hop	in binary
Engineering	200.23.16.0/23	uc-gw	11001000 00010111 0001 000...
Science	200.23.18.0/23	uc-gw	11001000 00010111 0001 001...
Law	200.23.20.0/23	uc-gw	11001000 00010111 0001 010...
Arts	200.23.22.0/23	uc-gw	11001000 00010111 0001 011...
...	...	uc-gw	...
Library	200.23.30.0/23	uc-gw	11001000 00010111 0001 111...

THE ANSWER

- The answer is **address aggregation**.
- In this example, all eight entries agree on their **first 20 bits** and share a next hop. So the eight can be replaced by one entry, **200.23.16.0/20**.

WHEN ONE NETWORK IS REACHED ANOTHER WAY

What if two entries both match?

A ROUTER'S FORWARDING TABLE

network	destination	next hop
the whole university	200.23.16.0/20	uc-gw
Science	200.23.18.0/23	cloud-gw

THE FORWARDING DECISION

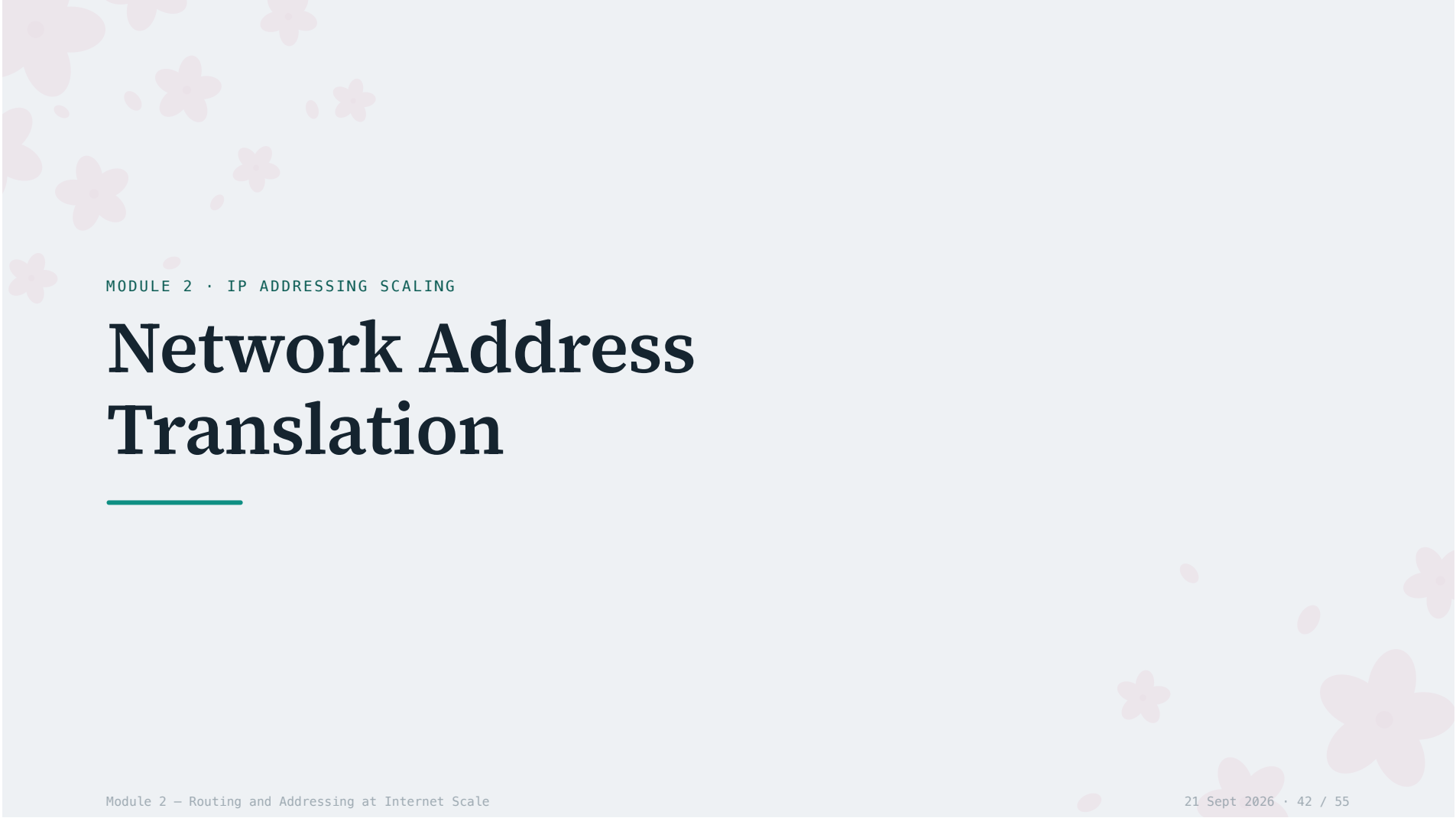
- The router holds that one aggregated entry. Now suppose **Science (200.23.18.0/23)** moves to a different provider, so its **next hop changes to cloud-gw**.
- As a result we add a **separate entry** for Science to this table.
- Now suppose a datagram arrives for **200.23.18.5**. It matches **both** destination prefixes — 20 bits on one, 23 on the other.
- **Which entry should the router use?**

LONGEST-PREFIX MATCH

When more than one entry matches a destination address, forward using the entry with the **longest** prefix.

Here 23 beats 20, so the datagram goes to **cloud-gw**.

Longest-match forwarding: RFC 4632 §5.1 · RFC 1812 §5.2.4.3



MODULE 2 • IP ADDRESSING SCALING

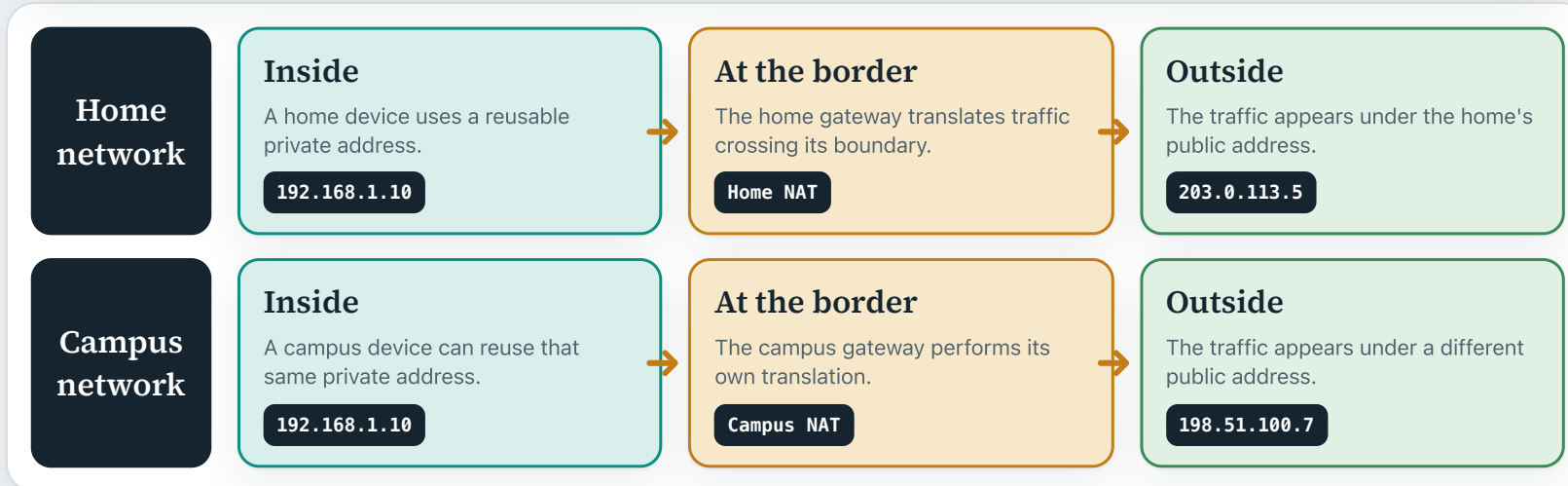
Network Address Translation

Why do so many local networks use addresses like 192.168.1.10?

- **A familiar pattern.** You may see addresses such as 192.168.1.10 on devices at home, in an office, or in a campus lab.
- **Why is there no conflict?** If both your home and a campus lab contain 192.168.1.10, why does traffic sent at home reach the home device rather than the lab device?

NAT lets many private hosts share scarce public IPv4 addresses

As a result, the outside world only ever observes those two public addresses — and there is **no conflict**.



NAT translates both the address and the port

Inside

1 • Private address

Suppose the home device has this private IPv4 address, assigned through **DHCP** (Dynamic Host Configuration Protocol).

192.168.1.10

2 • Source port

The device wants to send a request to the server, so it opens a **source port** for this request.

51514

3 • Original packet

From that address and port, it sends a packet to the destination server.

S 192.168.1.10:51514
D 198.51.100.20:443

At the border

4 • Public address

The home gateway has its own unique public IPv4 address, typically assigned by its ISP.

203.0.113.5

5 • Assign a public port

NAT chooses an available public port for this outbound connection and saves the mapping.

PRIVATE SOURCE		PUBLIC SOURCE
192.168.1.10:51514	→	203.0.113.5:40001
[address : port]		[address : port]

6 • Translated packet

S 203.0.113.5:40001
D 198.51.100.20:443

Outside

7 • Route to the server

The translated packet is then routed to the destination server.

198.51.100.20:443

How does the reply reach the correct home device?

Inside

4 · Deliver inside

The rewritten packet goes to the correct home device and port.

```
S 198.51.100.20:443
D 192.168.1.10:51514
```

At the border

2 · Look up the mapping

NAT observes the public destination and finds the associated private destination.

PUBLIC DESTINATION	PRIVATE DESTINATION
203.0.113.5:40001	→ 192.168.1.10:51514

3 · Rewrite the destination

NAT replaces the public destination with the private address and port.

```
S 198.51.100.20:443
D 192.168.1.10:51514
```

Outside

1 · Reply arrives

The server replies to the public source tuple it observed.

```
S 198.51.100.20:443
D 203.0.113.5:40001
```

Advantages and disadvantages of NAT

Advantages

Address conservation: many devices can share one or a few public IPv4 addresses.

Addressing independence: internal addresses can change without affecting the public Internet, and the public address can change without renumbering every internal host.

Disadvantages

Per-flow state: the gateway must remember active translations.

Inbound reachability: internal devices are not directly addressable from the public Internet. A mapping only comes into existence once the device has sent something out, so a reply can find its way back — nothing outside can start the conversation.

IPv6

- IPv6 is the **newer version** of the Internet Protocol.
- It uses **128-bit addresses**, far more than IPv4's 32-bit format.

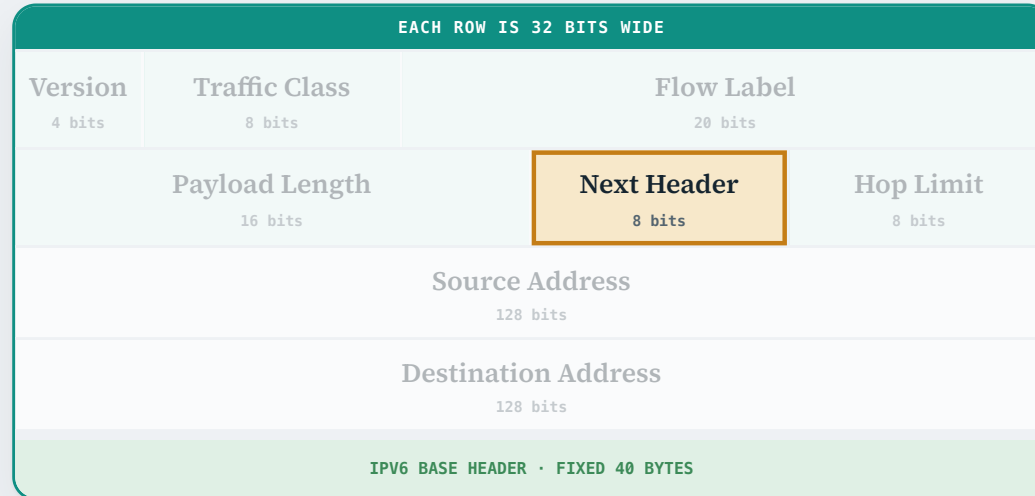
IPv6 address

- **First 48 bits • global routing prefix:** identify an allocated site or organization block. For example, a university could be allocated this block.
- **Next 16 bits • subnet ID:** identify one subnet inside that site. For example, this could be the Engineering department's network.
- **Final 64 bits • interface identifier:** identify one interface within that subnet. For example, this could be a laptop's Wi-Fi interface.



A breakdown of the IPv6 header

Every IPv6 packet begins with the same **fixed 40-byte base header**.



Source and destination addresses

Both fields are 128 bits and identify the packet's endpoints.

Hop Limit

Each router reduces it by one so a packet caught in a forwarding loop is eventually discarded.

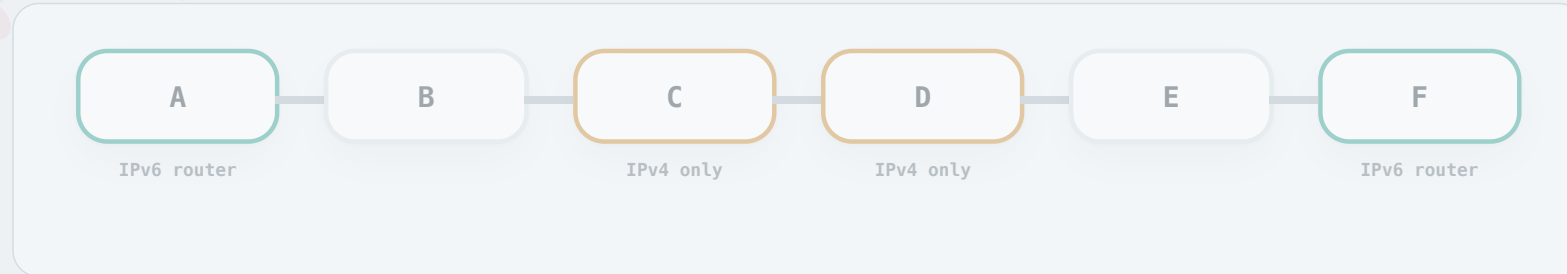
Flow Label

Marks packets belonging to the same flow — a TCP connection, for example — so routers can treat them alike.

Next Header

Identifies the extension header or upper-layer protocol that follows, such as a TCP or UDP header.

IPv6 traffic may encounter IPv4-only routers



- IPv6 is still being deployed, and progress has been slow: many routers in service are older commercial devices that support only IPv4.
- Suppose IPv6 router A wants to send an IPv6 packet to IPv6 router F.
- The path crosses IPv4-only routers C and D, which cannot forward that packet directly.
- **How can we resolve this compatibility problem?**

Two ways across the IPv4-only region



Strategy 2 · Tunnelling:

- B **encapsulates** the entire IPv6 packet — header and payload together — into the **data field** of a new IPv4 packet.
- B then forwards that IPv4 packet to its next hop, C.



Module 2 recap

- **Intra-AS routing:** RIP uses distance vector, while OSPF uses link state and areas to route within an AS.
- **Inter-AS routing:** BGP learns AS-level paths, using eBGP between ASes and iBGP within an AS.
- **IP addressing:** NAT conserves public IPv4 addresses, and IPv6 expands the address space.

References

William Stallings

Data and Computer Communications, 10th edition, Pearson, 2013 · Chapter 19, Routing.

Andrew S. Tanenbaum & David J. Wetherall

Computer Networks, 5th edition, Prentice Hall, 2010 · Chapter 5, The Network Layer.

James F. Kurose & Keith W. Ross

Computer Networking: A Top-Down Approach Featuring the Internet, 3rd edition, Addison Wesley, 2005 · Chapter 4, The Network Layer.

Core Internet standards

RFC 4271, 4632, 5737, 6598, 6888, 8200, and 4213, together with the protocol-specific RFCs cited on individual frames.

Number-resource authorities

IANA Number Resources and APNIC registry, allocation, and address-management material.

Teaching examples

Teaching topologies, addresses, and routing policies are explicitly constructed examples unless a frame identifies an external source.

Standards and registry sources are identified on the frames where they are used.

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